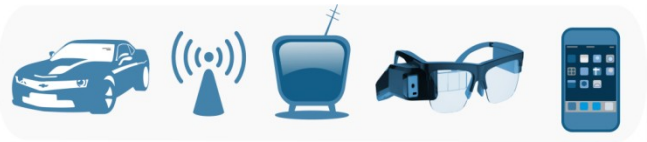
EUROGICIEL Groupe



Porting Tizen IVI to Wayland







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LPC2014

EUROGICIEL GROUPE

Eurogiciel



Open-source development and integration :

- Maintainers for tizen.org (Base, Test, Web Framework, … domains)
- Embedded systems for real-time multimedia :
  - Widi/Miracast stack,
  - Wayland/Weston,
  - Webkit2 browser with HW acceleration.
- Applications : HTML5/CSS3, jquery, jqmobi, Cordova
- Location : Brittany - France



















About me

Manuel Bachmann

<manuel.bachmann@open.eurogiciel.org>



- Previous maintainer of the GTK+3 for Win32 packages ;
- Author of a GLX wrapper for EGL : EGLX [(1)] ;
- Upstream contributor to Weston, EFL, Ozone-Wayland... ;
- Maintains Weston and UI domain packages under Tizen.


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Summary

- IVI : In-Vehicle Infotainment
- Making Tizen IVI Wayland-only ;
- Crosswalk web framework, Tizen APIs and XDG-Shell ;
- IVI-Shell for native extended IVI capabilities ;
- Frameworks and toolkits integration.







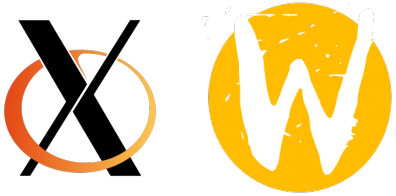
IVI : In-Vehicle Infotainment

- dedicated to vehicles ;
- following the GENIVI (1) alliance specification ;
- two main flavors :
  - Modello Homescreen with the Crosswalk web framework ;
  - ICO Homescreen with IVI-Shell.

<DEMO TIME>

- 1) Tizen IVI with Modello Homescreen
- 2) Tizen IVI with ICO Homescreen

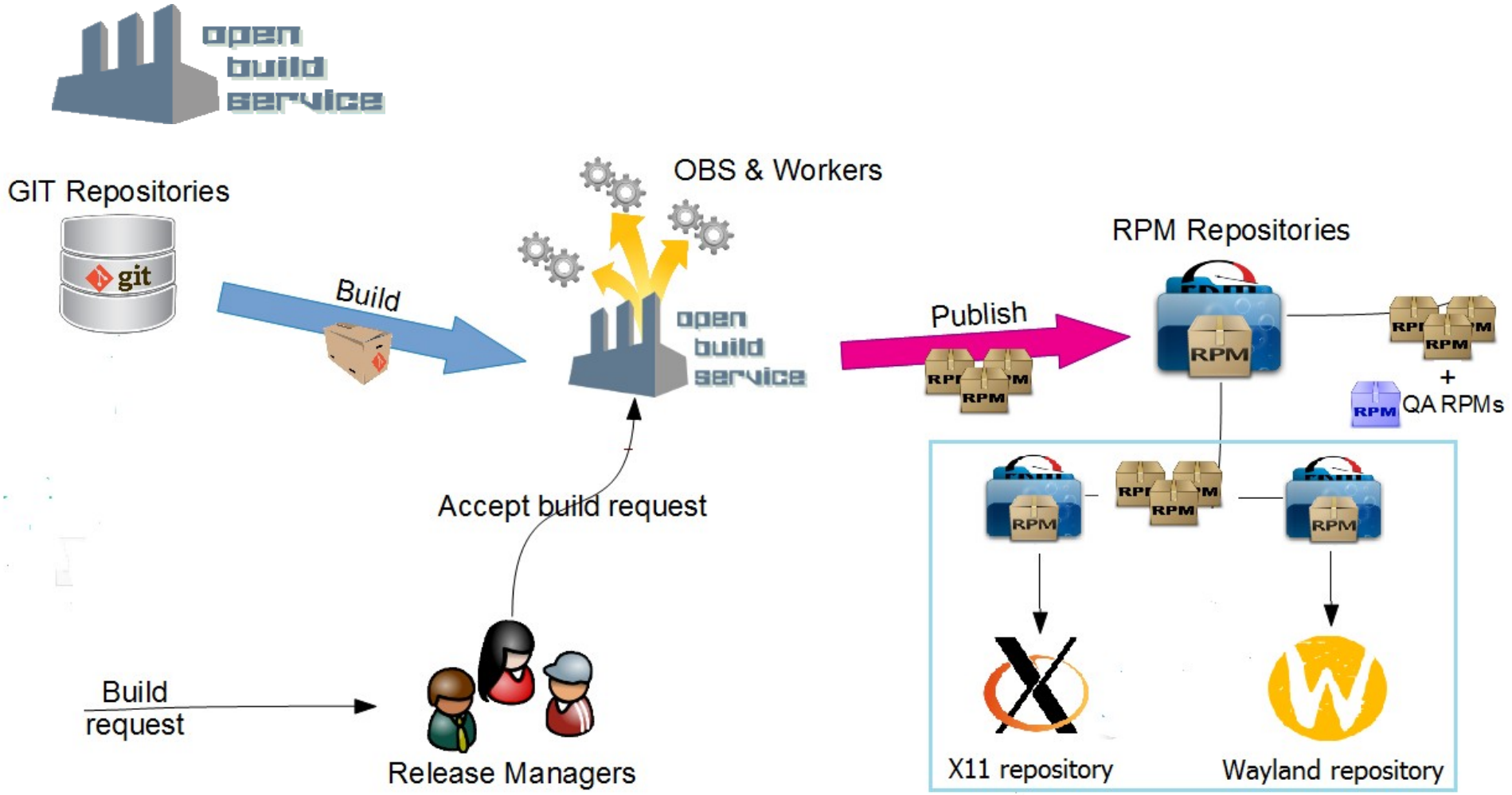




Making Tizen IVI Wayland-only

- Tizen Common supports several architectures (X86, AMD64, ARM…) and, for each one, a X11 and a Wayland profile ;
- Macros (%with_x, %with_wayland, ...) ensure building all packages for the right profile ;
- Tizen IVI is a x86 (Atom) Wayland profile with specifics ;
- Cleaned out of its X11 crufts by April 2014, thanks to release manager Mikko Ylinen.

Making Tizen IVI Wayland-only



from Git repositories to X11/Wayland packages





- Based on Chromium, with specifics in the web engine and additional Tizen extensions ;
- Ozone-Wayland backend uses Wayland XDG-Shell for some Tizen API functionalities :
    - *tizen.application.hide()* is mapped to *xdg_surface_set_minimized()* ;
    - *tizen.application.launch()*, on an already started application, will resume it and bring it to foreground with *xdg_surface_present()* (Tizen patch)*;*
- Hope to see *xdg_surface_present()* in the next Weston upstream release [(2)].

<DEMO TIME>

- 1) demo app with hide()/launch()
- 2) Tizen IVI - Modello Homescreen and hide()/launch()





IVI-Shell for native extended IVI capabilities

- In review process upstream, developer is TANIBATA Nobuhiko, hope to see it in Weston 1.7.0 ;
- Provides a Weston shell interface compliant with the GENIVI specification [(1)] ;
- We maintain an external Weston Git repository with patches applied for reviewers [(3)], and build it as a separate plugin under Tizen ;
- Compatibility layer with XDG-Shell, so that standard applications can use it, has been proven possible and is currently being done [(4)].

<DEMO TIME>

- 1) Tizen IVI - IVI-Shell and XDG-Shell compat demo









Frameworks and toolkits integration

- EFL : XDG-Shell and IVI-Shell pushed and integrated upstream [5] ;
- Qt : XDG-Shell pushed and integrated upstream by Philippe Coval, IVI-Shell under review [6] ;
- Crosswalk/Ozone-Wayland : XDG-Shell and IVI-Shell pushed and integrated upstream [7].

Links



- (1) : GENIVI alliance : http://www.genivi.org
- (2) : xdg_surface_present() implementation demo : http://lists.freedesktop.org/archives/wayland-devel/2014-July/016224.html
- (3) : weston-ivi-shell upstream adaptation GitHub repository : https://github.com/Tarnyko/weston-ivi-shell
- (4) : ivi-shell xdg-shell compatibility layer feasability : https://www.mail-archive.com/ivi@lists.tizen.org/msg02702.html
- (5) : EFL upstream wayland shells : http://git.enlightenment.org/core/efl.git/commit/?id=87f02170e659678d7a2f000e6850bd3a29962756
  - https://git.enlightenment.org/core/efl.git/commit/?id=50287ab731d4d87170238b365203e830edc038d5

Links (2) 

- [6] : Qt upstream wayland shells : https://bugreports.qt-project.org/browse/QTBUG-38633 - https://bugreports.qt-project.org/browse/QTBUG-41172
- [7] : Ozone-Wayland upstream wayland shells : https://github.com/01org/ozone-wayland/commit/5f8a34c613ba826c7994c81d03f87df19f48881d - https://github.com/01org/ozone-wayland/commit/a034a018b6ec317ec5559dcce6efec916ec40512

Q&A

That's all folks !

